TP- implement single-area ospfv2

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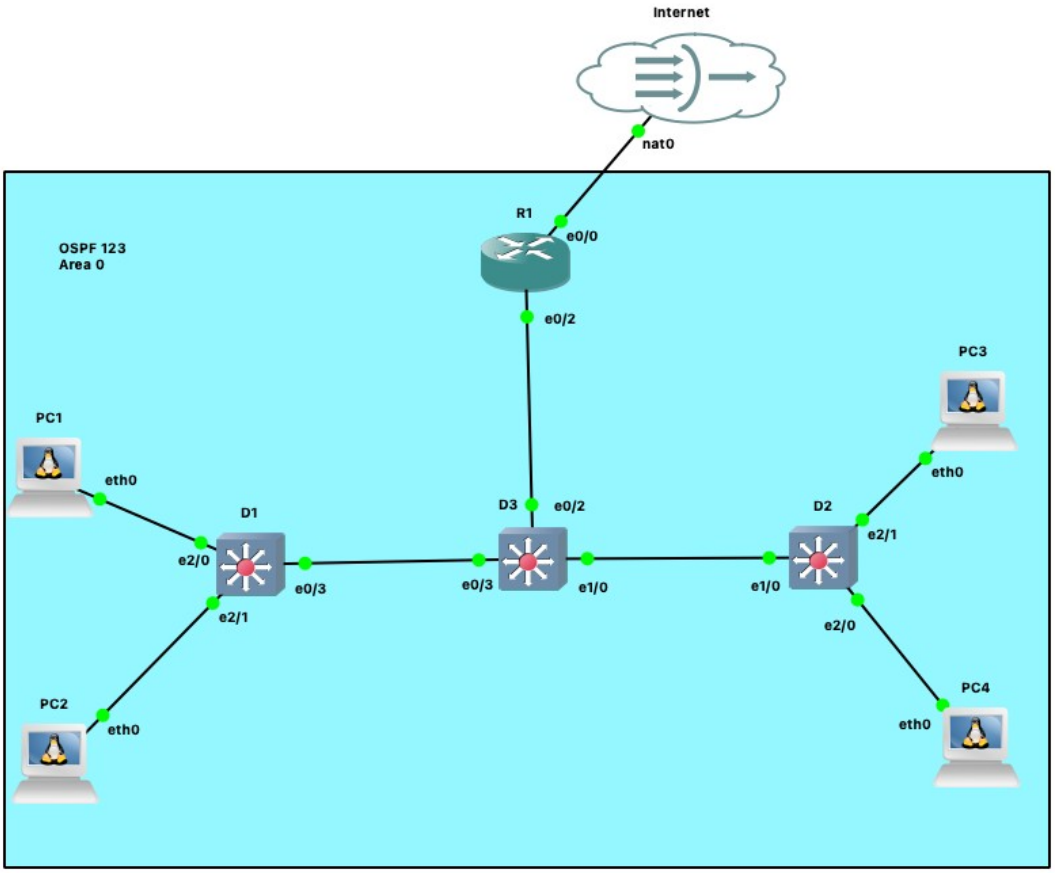
Topology




```
Bmarco11 — R1 — telnet 192.168.71.128 5003 — 80×24
            ^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
*Nov  7 12:20:24.095: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief | include manual
Ethernet0/2            10.10.0.1       YES manual up                    up
Loopback0              209.165.200.225 YES manual up                    up
Loopback1              192.168.1.1     YES manual up                    up
R1#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 80×24

```
D1(config)#interface e0/3
D1(config-if)#no switchport
D1(config-if)#ip add 10.10.0.2 255.255.255.248
D1(config-if)#no shutdown
D1(config-if)#exit
*Nov  7 12:33:20.414: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to up
*Nov  7 12:33:21.414: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to up
D1(config-if)#exit
D1(config)#end
D1#
*Nov  7 12:33:31.912: %SYS-5-CONFIG_I: Configured from console by console
D1#show ip interface brief | include manual
Ethernet0/3            10.10.0.2       YES manual up                    up
Ethernet2/0            10.10.8.1       YES manual up                    up
Ethernet2/1            10.10.9.1       YES manual up                    up
D1#
```

Bmarco11 — D2 — telnet 192.168.71.128 5002 — 80×24

```
D2(config-if)#ip address 10.10.24.1 255.255.255.0
D2(config-if)#no shutdown
D2(config-if)#exit
D2(config)#inerface e2/0
             ^
% Invalid input detected at '^' marker.

D2(config)#interface e2/0
D2(config-if)#no switchport
D2(config-if)#
*Nov  7 12:19:27.292: %LINK-3-UPDOWN: Interface Ethernet2/0, changed state to up
*Nov  7 12:19:28.297: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/
0, changed state to up
D2(config-if)#ip address 10.10.25.1 255.255.255.0
D2(config-if)#no shutdown
D2(config-if)#exit
D2(config)#exit
D2#show ip interface brief | include manual
*Nov  7 12:22:19.839: %SYS-5-CONFIG_I: Configured from console by console
D2#show ip interface brief | include manual
Ethernet1/0            10.10.0.3       YES manual up                    up
Ethernet2/0            10.10.25.1      YES manual up                    up
Ethernet2/1            10.10.24.1      YES manual up                    up
D2#
```

```
R1#
R1#ping 10.10.0.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.0.3, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/4 ms
R1#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 80×17

```
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 123"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.10.9.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.0.2 0.0.0.0 area 0
    10.10.8.1 0.0.0.0 area 0
    10.10.9.1 0.0.0.0 area 0
 --More--
```

Bmarco11 — D2 — telnet 192.168.71.128 5002 — 80×24

```
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 123"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.10.25.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.0.0 0.0.0.7 area 0
    10.10.24.0 0.0.0.255 area 0
    10.10.25.0 0.0.0.255 area 0
 --More--
```

Bmarco11 — R1 — telnet 192.168.71.128 5003 — 80×19

```
R1#
*Nov  7 12:43:30.383: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.9.1 on Ethernet0/2
from LOADING to FULL, Loading Done
R1#show ip protocols | section ospf
Routing Protocol is "ospf 123"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 209.165.200.225
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Routing on Interfaces Configured Explicitly (Area 0):
    Loopback1
    Ethernet0/2
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.9.1            110      00:00:06
    10.10.25.1           110      00:00:06
R1#
```

Bmarco11 — R1 — telnet 192.168.71.128 5003 — 80×19

```
*Nov  7 12:46:46.535: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.9.1 on Ethernet0/2
from FULL to DOWN, Neighbor Down: Interface down or detached
*Nov  7 12:46:46.535: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.25.1 on Ethernet0/2
 from FULL to DOWN, Neighbor Down: Interface down or detached
*Nov  7 12:46:46.543: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.9.1 on Ethernet0/2
from LOADING to FULL, Loading Done
*Nov  7 12:46:46.543: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.25.1 on Ethernet0/2
 from LOADING to FULL, Loading Done
R1#show ip protocol | include Router ID
  Router ID 1.1.1.1
R1#show ip route ospf | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
O        10.10.8.0/24 [110/20] via 10.10.0.2, 00:00:44, Ethernet0/2
O        10.10.9.0/24 [110/20] via 10.10.0.2, 00:00:44, Ethernet0/2
O        10.10.24.0/24 [110/20] via 10.10.0.3, 00:00:44, Ethernet0/2
O        10.10.25.0/24 [110/20] via 10.10.0.3, 00:00:44, Ethernet0/2
R1#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 80×17

```
*Nov  7 12:47:27.133: %OSPF-5-ADJCHG: Process 123, Nbr 209.165.200.225 on Ethern
et0/3 from FULL to DOWN, Neighbor Down: Dead timer expired
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

D1#
D1#
D1#show ip route ospf | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
O        10.10.24.0/24 [110/20] via 10.10.0.3, 00:07:42, Ethernet0/3
O        10.10.25.0/24 [110/20] via 10.10.0.3, 00:07:42, Ethernet0/3
      192.168.1.0/26 is subnetted, 1 subnets
O        192.168.1.0 [110/11] via 10.10.0.1, 00:01:31, Ethernet0/3
D1#
```

Bmarco11 — D2 — telnet 192.168.71.128 5002 — 80×24

```
    10.10.25.0 0.0.0.255 area 0
 --More--
*Nov  7 12:43:26.466: %OSPF-5-ADJCHG: Process 123, Nbr 209.165.200.225 on Ethern
et1/0 from LOADING to FULL, Loading Done
 --More--
*Nov  7 12:46:47.177: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.1 on Ethernet1/0 fr
om LOADING to FULL, Loading Done
 --More--
*Nov  7 12:47:27.139: %OSPF-5-ADJCHG: Process 123, Nbr 209.165.200.225 on Ethern
et1/0 from FULL to DOWN, Neighbor Down: Dead timer expired
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.9.1            110      00:00:23
  Distance: (default is 110)

D2#show ip route ospf | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
O        10.10.8.0/24 [110/20] via 10.10.0.2, 00:08:32, Ethernet1/0
O        10.10.9.0/24 [110/20] via 10.10.0.2, 00:08:32, Ethernet1/0
      192.168.1.0/26 is subnetted, 1 subnets
O        192.168.1.0 [110/11] via 10.10.0.1, 00:02:16, Ethernet1/0
D2#
```

```
R1#show ip ospf interface brief
Interface     PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Lo1           123   0               192.168.1.1/26     1     P2P   0/0
Et0/2         123   0               10.10.0.1/29       10    DROTH 2/2
R1#
```

```
D1#show ip ospf interface brief
Interface     PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Et2/1         123   0               10.10.9.1/24       10    DR    0/0
Et2/0         123   0               10.10.8.1/24       10    DR    0/0
Et0/3         123   0               10.10.0.2/29       10    DR    2/2
D1#
```

```
O        192.168.1.0 [110/11] via 10.10.0.1, 00:02:16, Ethernet1/0
D2#show ip ospf interface brief
Interface     PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Et2/0         123   0               10.10.25.1/24      10    DR    0/0
Et2/1         123   0               10.10.24.1/24      10    DR    0/0
Et1/0         123   0               10.10.0.3/29       10    BDR   2/2
D2#
```

Bmarco11 — R1 — telnet 192.168.71.128 5003 — 80×19

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.10.9.1         1   FULL/DR         00:00:33    10.10.0.2       Ethernet0/2
10.10.25.1        1   FULL/BDR        00:00:37    10.10.0.3       Ethernet0/2
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 0.0.0.0 0.0.0.0 lo0
%Default route without gateway, if not a point-to-point interface, may impact pe
rformance
R1(config)#router ospf 123
R1(config-router)#default-information originate
R1(config-router)#end
R1#
*Nov  7 12:52:34.883: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route static | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

S*    0.0.0.0/0 is directly connected, Loopback0
R1#
```

```
D2#show ip route | include Gateway|0/0
Gateway of last resort is 10.10.0.1 to network 0.0.0.0
O*E2  0.0.0.0/0 [110/1] via 10.10.0.1, 00:01:01, Ethernet1/0
D2#
```

```
R1#
*Nov  7 13:00:32.147: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip ospf | include Ref
 Reference bandwidth unit is 1000 mbps
R1#
```

Bmarco11 — R1 — telnet 192.168.71.128 5003 — 80×19

```
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 123
R1(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
R1(config-router)#end
R1#
*Nov  7 13:00:32.147: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip ospf | include Ref
 Reference bandwidth unit is 1000 mbps
R1#show ip route ospf | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
O        10.10.8.0/24 [110/200] via 10.10.0.2, 00:00:49, Ethernet0/2
O        10.10.9.0/24 [110/200] via 10.10.0.2, 00:00:49, Ethernet0/2
O        10.10.24.0/24 [110/200] via 10.10.0.3, 00:00:18, Ethernet0/2
O        10.10.25.0/24 [110/200] via 10.10.0.3, 00:00:18, Ethernet0/2
R1#
```

Bmarco11 — R1 — telnet 192.168.71.128 5003 — 80×19

```
*Nov  7 13:06:11.047: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.25.1 on Ethernet0/2
 from FULL to DOWN, Neighbor Down: Dead timer expired
R1#show ip ospf interface e0/2 | include Timer
*Nov  7 13:06:33.727: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.9.1 on Ethernet0/2
from FULL to DOWN, Neighbor Down: Dead timer expired
R1#show ip ospf interface e0/2 | include Timer
  Timer intervals configured, Hello 5, Dead 20, Wait 20, Retransmit 5
R1#
*Nov  7 13:09:27.655: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.9.1 on Ethernet0/2
from LOADING to FULL, Loading Done
R1#
*Nov  7 13:11:23.031: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.25.1 on Ethernet0/2
 from LOADING to FULL, Loading Done
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.10.9.1         1   FULL/BDR        00:00:16    10.10.0.2       Ethernet0/2
10.10.25.1        1   FULL/DR         00:00:18    10.10.0.3       Ethernet0/2
R1#
```

Bmarco11 — D1 — telnet 192.168.71.128 5000 — 80×17

```
D1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)# interface e0/3
D1(config-if)#ip ospf hello-interval 5
D1(config-if)#
*Nov  7 13:09:28.290: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.1 on Ethernet0/3 fr
om LOADING to FULL, Loading Done
D1(config-if)#ip ospf dead-interval 20
D1(config-if)#end
D1#
*Nov  7 13:09:40.863: %SYS-5-CONFIG_I: Configured from console by console
D1#
*Nov  7 13:09:56.267: %OSPF-5-ADJCHG: Process 123, Nbr 10.10.25.1 on Ethernet0/3
 from FULL to DOWN, Neighbor Down: Dead timer expired
D1#show ip ospf interface e0/3 | include Timer
  Timer intervals configured, Hello 5, Dead 20, Wait 20, Retransmit 5
D1#
```

```
D2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:16    10.10.0.1       Ethernet1/0
10.10.9.1         1   FULL/DR         00:00:15    10.10.0.2       Ethernet1/0
D2#
```

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.10.9.1         1   FULL/DR         00:00:15    10.10.0.2       Ethernet0/2
10.10.25.1        0   FULL/DROTHER    00:00:18    10.10.0.3       Ethernet0/2
R1#
```